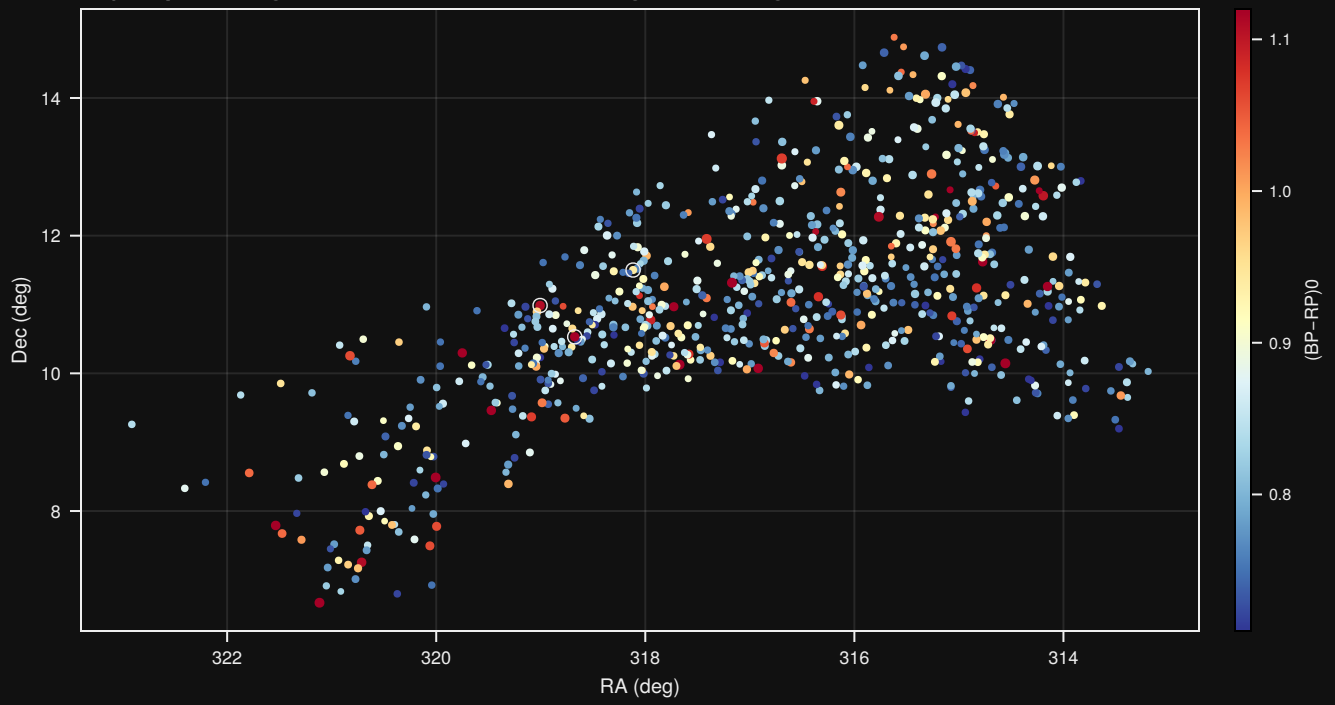
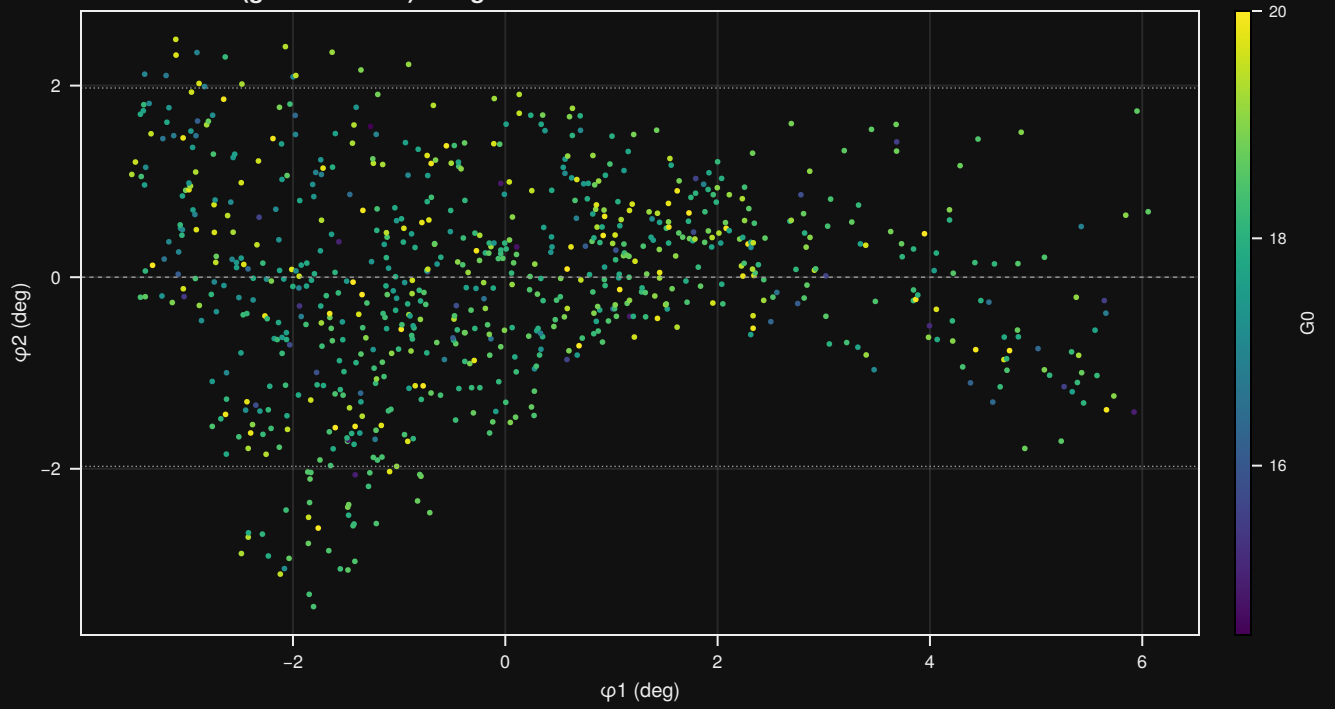
New-21

MEMBERSHIP	
Stars in catalog	762
with Gaia GSP-Phot	536 (70%)
with radial velocity	3
Stream id (sID)	74
GEOMETRY · great-circle fit	
Centroid RA, Dec	316.9, 11.0 deg
Centroid l, b	60.3, -23.3 deg
Pole RA, Dec	63.4, 55.3 deg
Length (φ1, 2–98%)	8.6 deg
Width σ(φ2), robust	0.99 deg
KINEMATICS	
pmRA* med ± σ	-2.90 ± 1.35 mas/yr
pmDE med ± σ	-2.64 ± 0.78 mas/yr
μφ1* med ± σ	-0.86 ± 1.21 mas/yr
μφ2 med ± σ	-3.75 ± 1.17 mas/yr
Parallax (cat.) med	0.211 ± 0.199 mas
Parallax (wtd, ZP-corr.)	0.220 ± 0.011 mas
Distance	d = 4.5 ± 0.2 kpc
Physical length	0.68 kpc
RV median ± σ(MAD)	-398.9 ± 107.7 km/s
RV gradient dV/dφ1	—
PHOTOMETRY · POPULATIONS	
G0 range (median)	14.5–20.0 (18.3)
(BP−RP)0 median	0.84
GSP-Phot [M/H] (n=536)	-1.21 ± 0.54
GSP-Phot Teff median	5270 K
RUWE > 1.4	0.4 %
PM clusters (largest)	49 (30 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	762 | 762
n_RV catalog | paper	3 | 3
<[Fe/H]> paper	-0.87
Stellar mass (paper)	7000 Msun
Pole offset vs paper	1.1 deg
Remark	High velocity

Sky map · catalog members (n = 762) · size ∝ brightness · ring = has RV

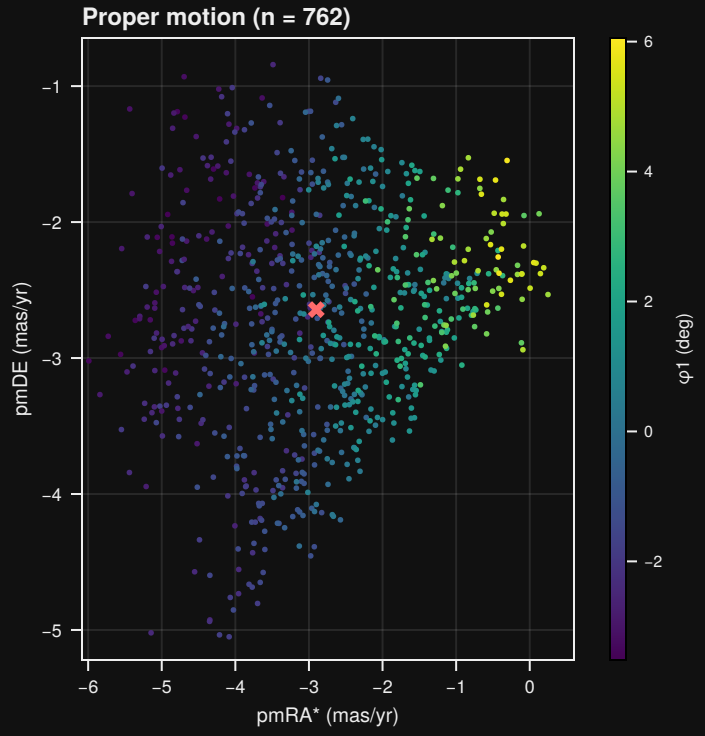


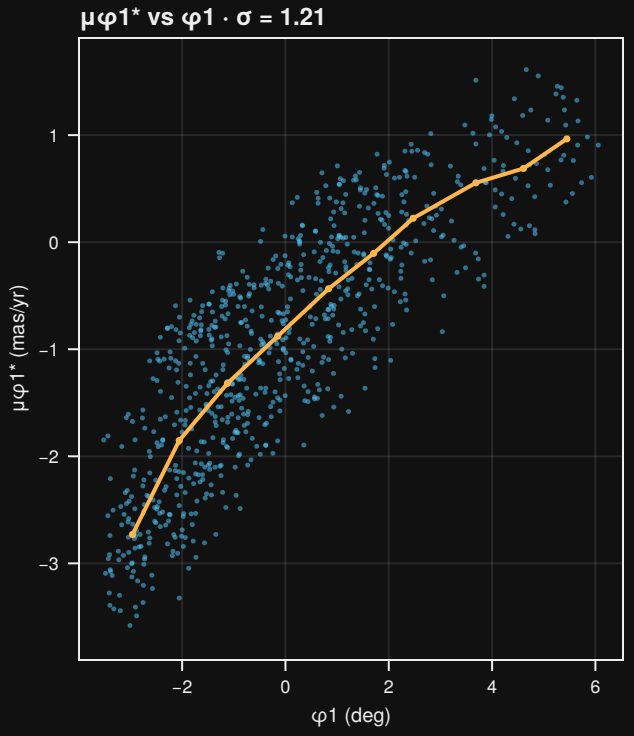
Stream frame (great-circle fit) · length 8.6° · width σ 0.99°

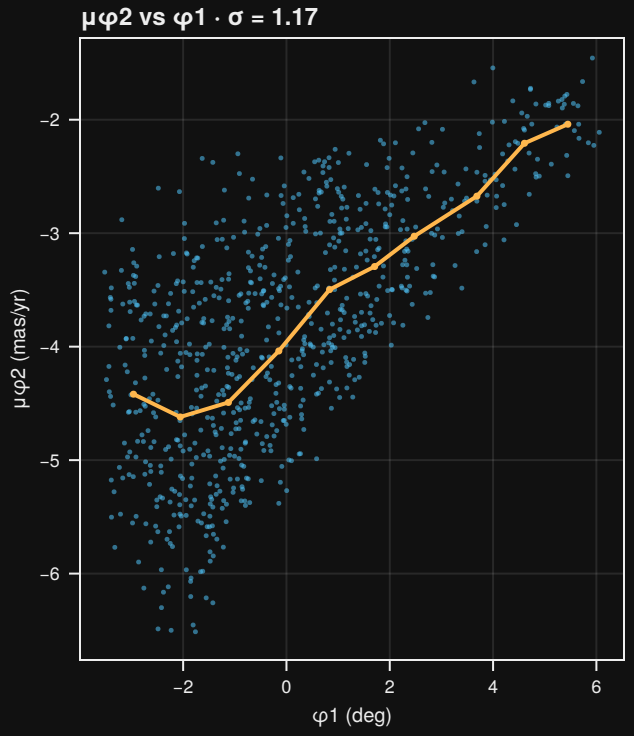


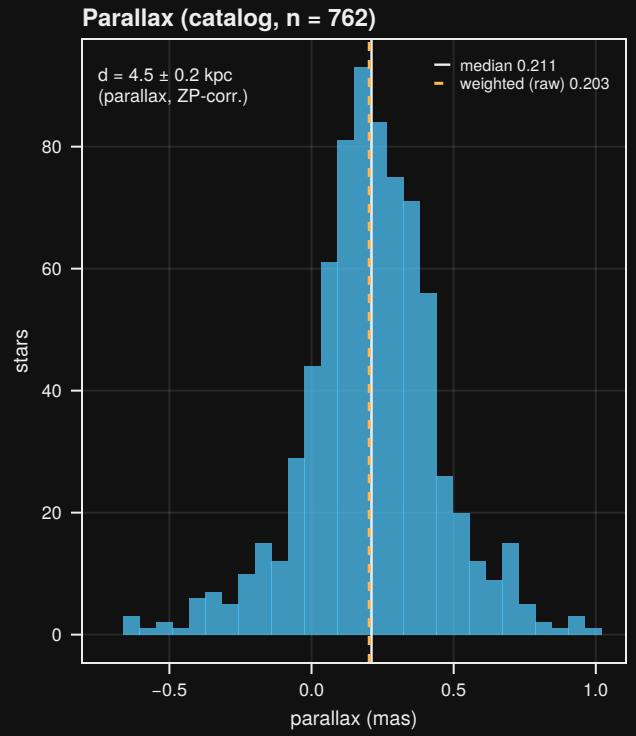
New-21

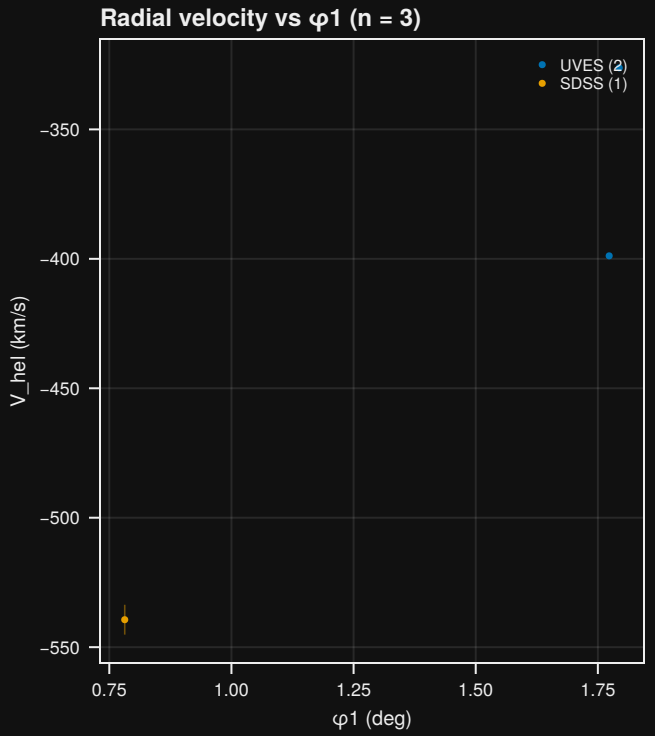
Kinematics · proper motion · parallax · radial velocity

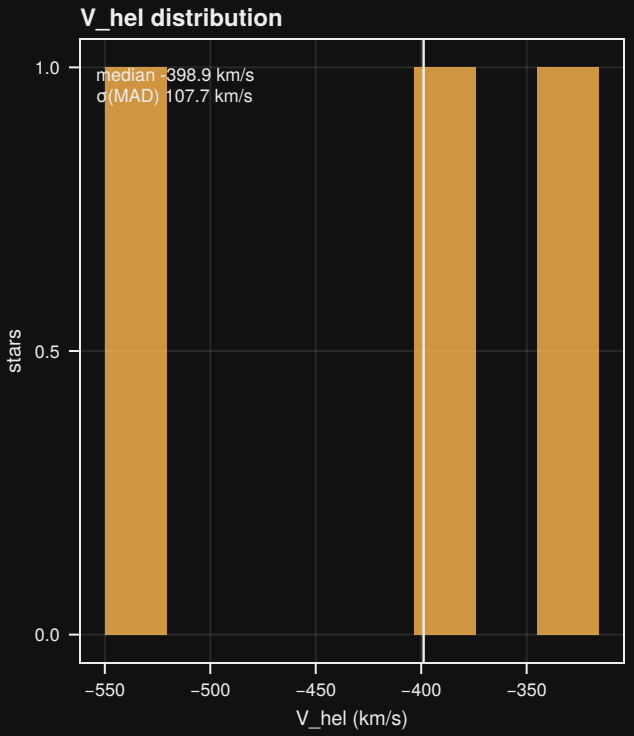






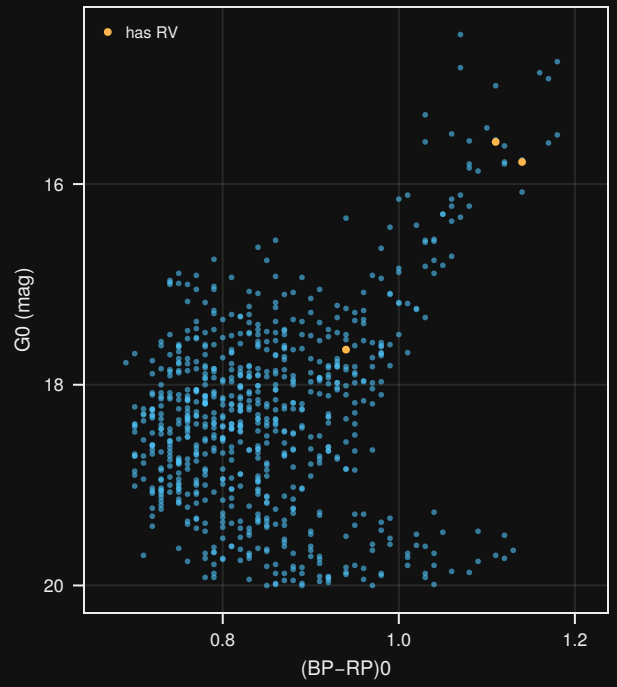




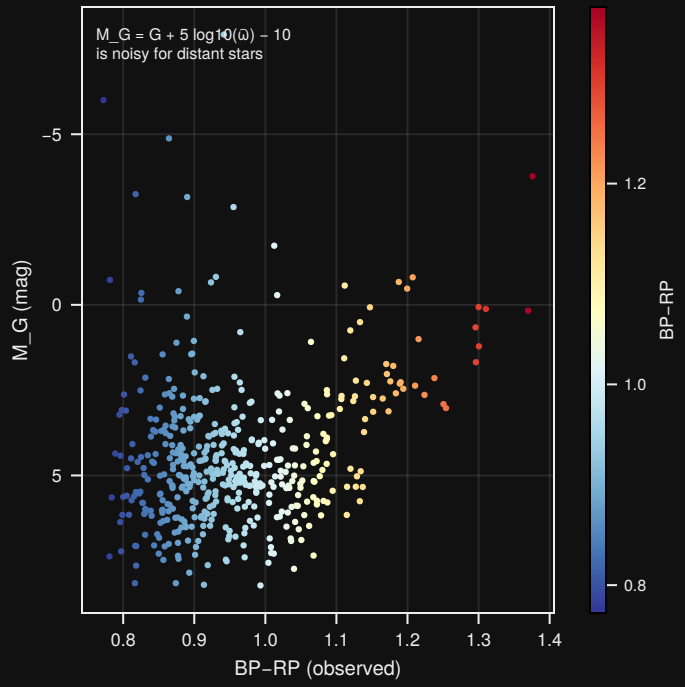


New-21

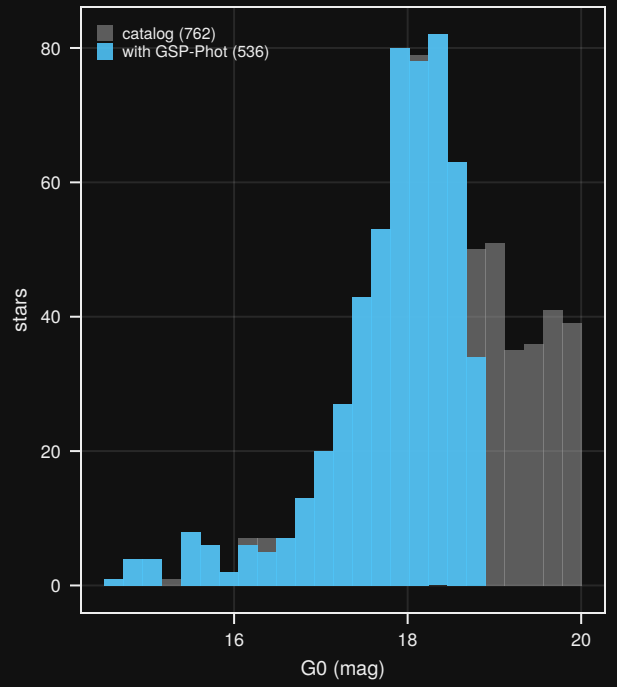
CMD · dereddened (n = 762)



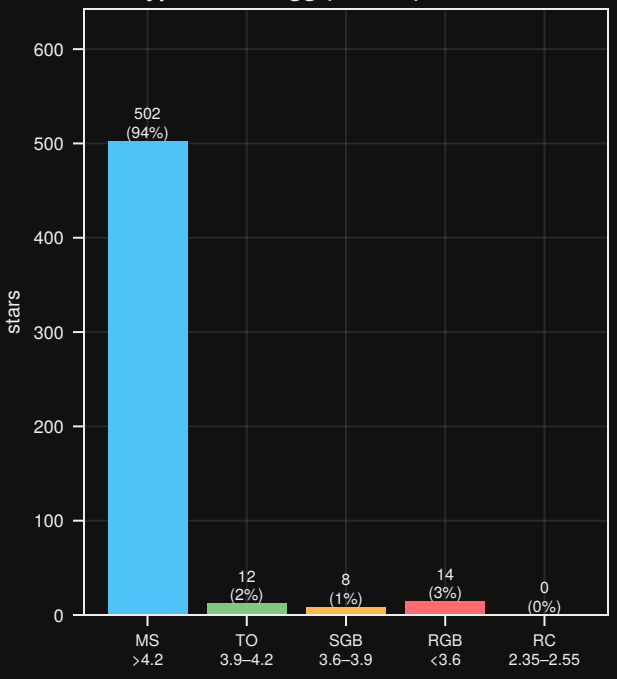
HR diagram · ϖ>0 (n = 483/536)



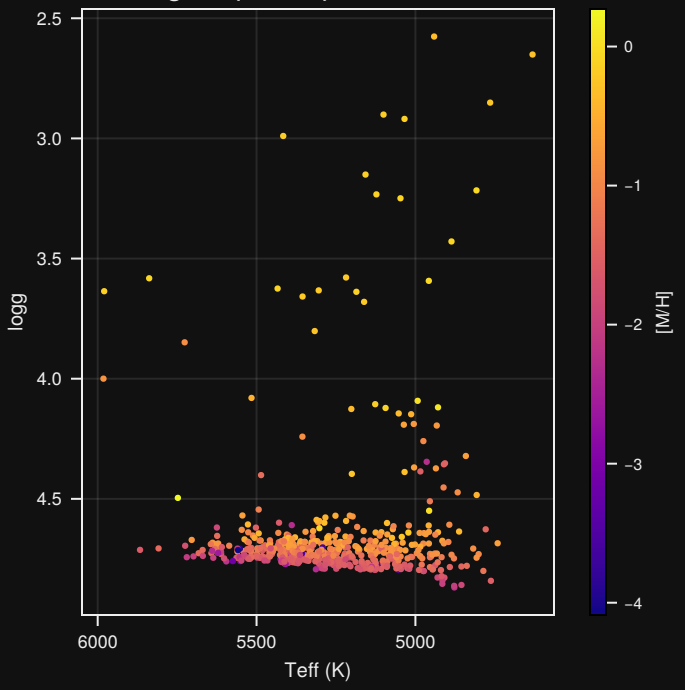
G0 & GSP-Phot coverage



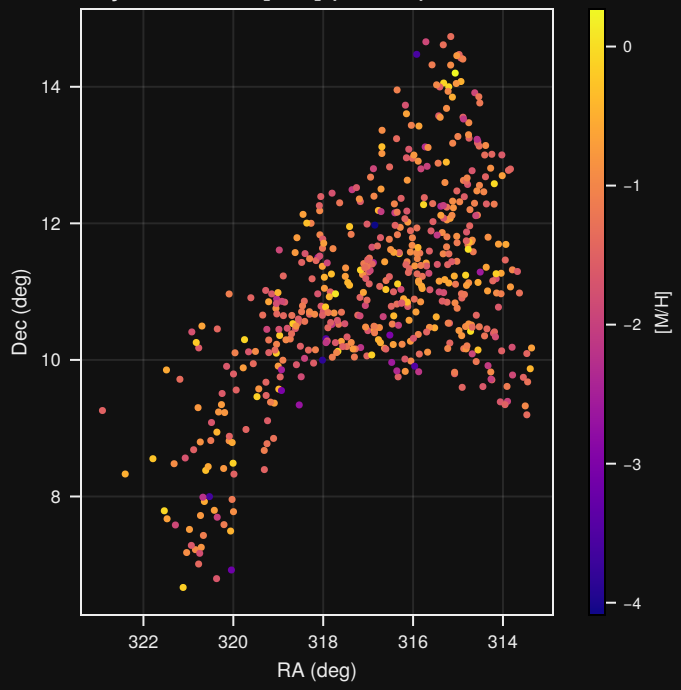
Star types from logg (n = 536)



Kiel diagram (n = 536)



Sky · GSP-Phot [M/H] (n = 536)

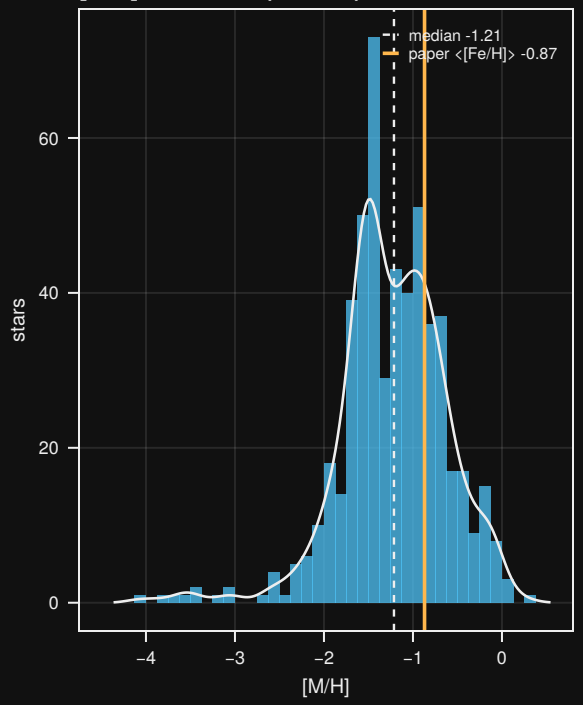


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

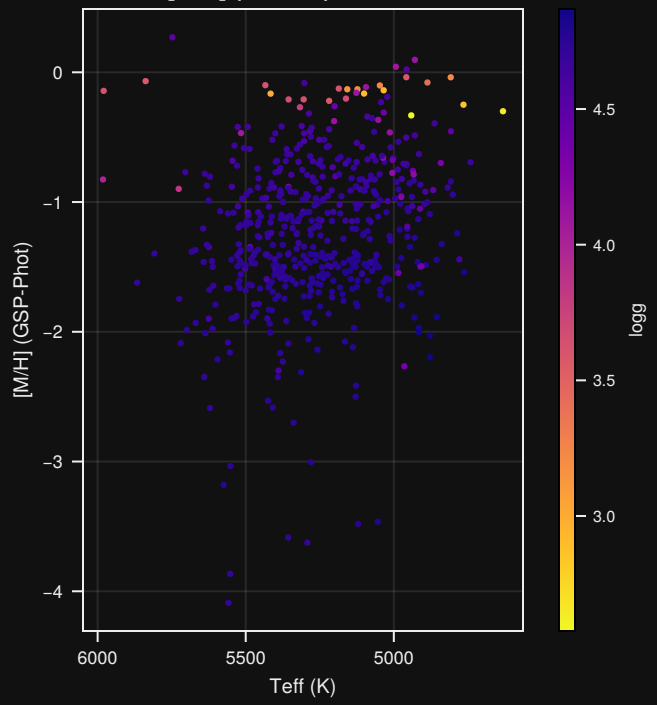
New-21

Metallicity · age · astrometric quality

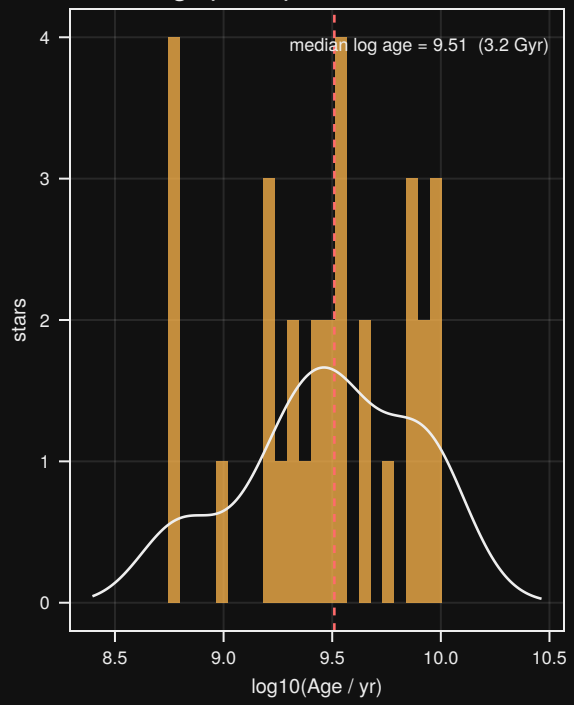
[M/H] GSP-Phot (n = 536)



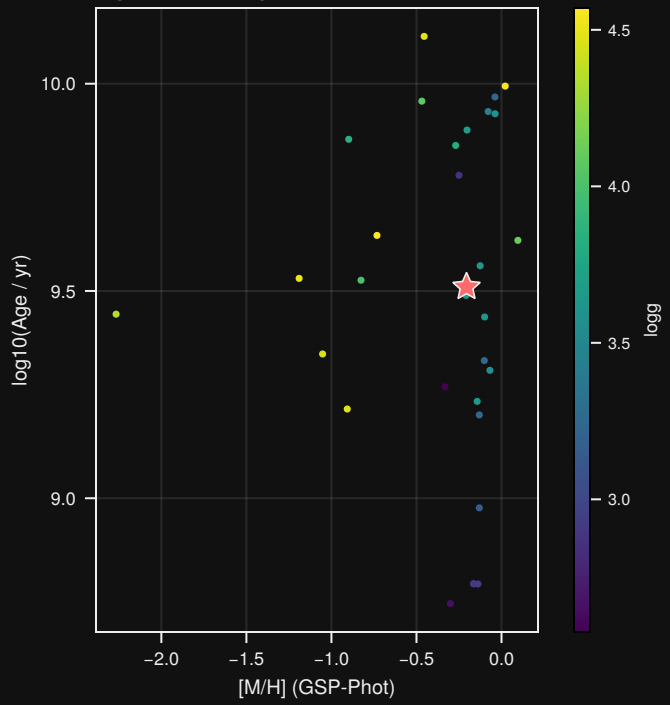
Teff vs [M/H] (n = 536)



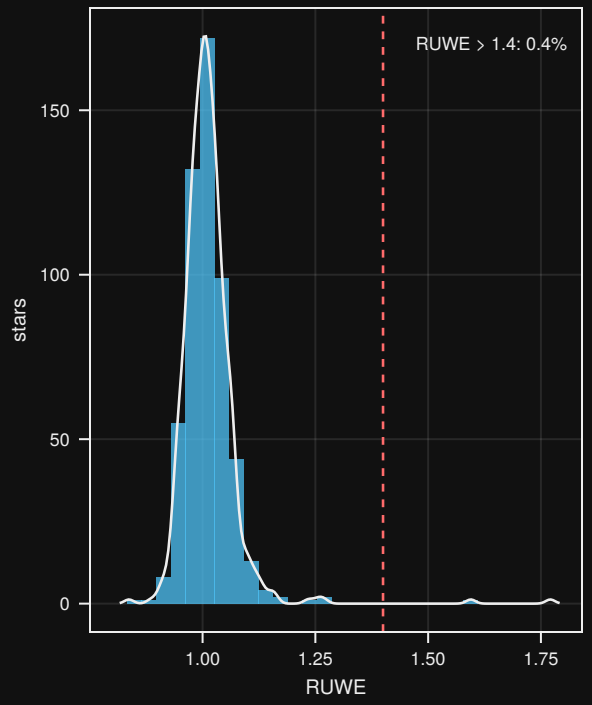
FLAME age (n = 32)



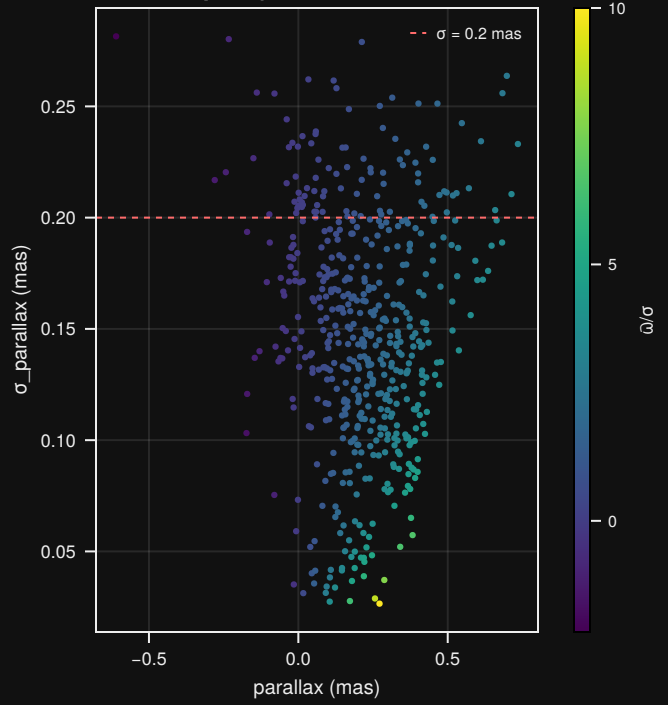
Age–metallicity (n = 32)



RUWE (n = 536)

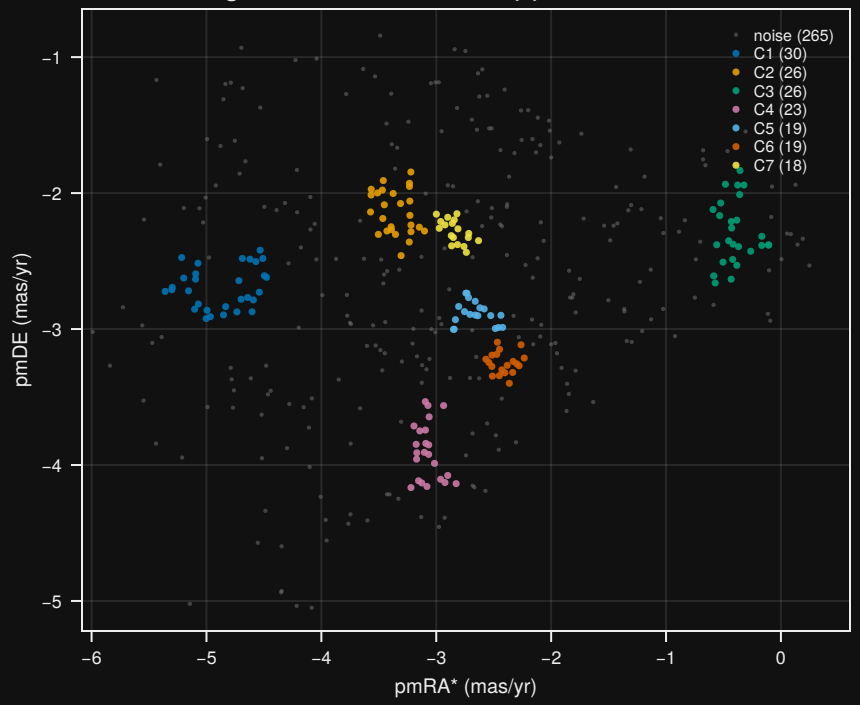


Parallax quality (n = 536)

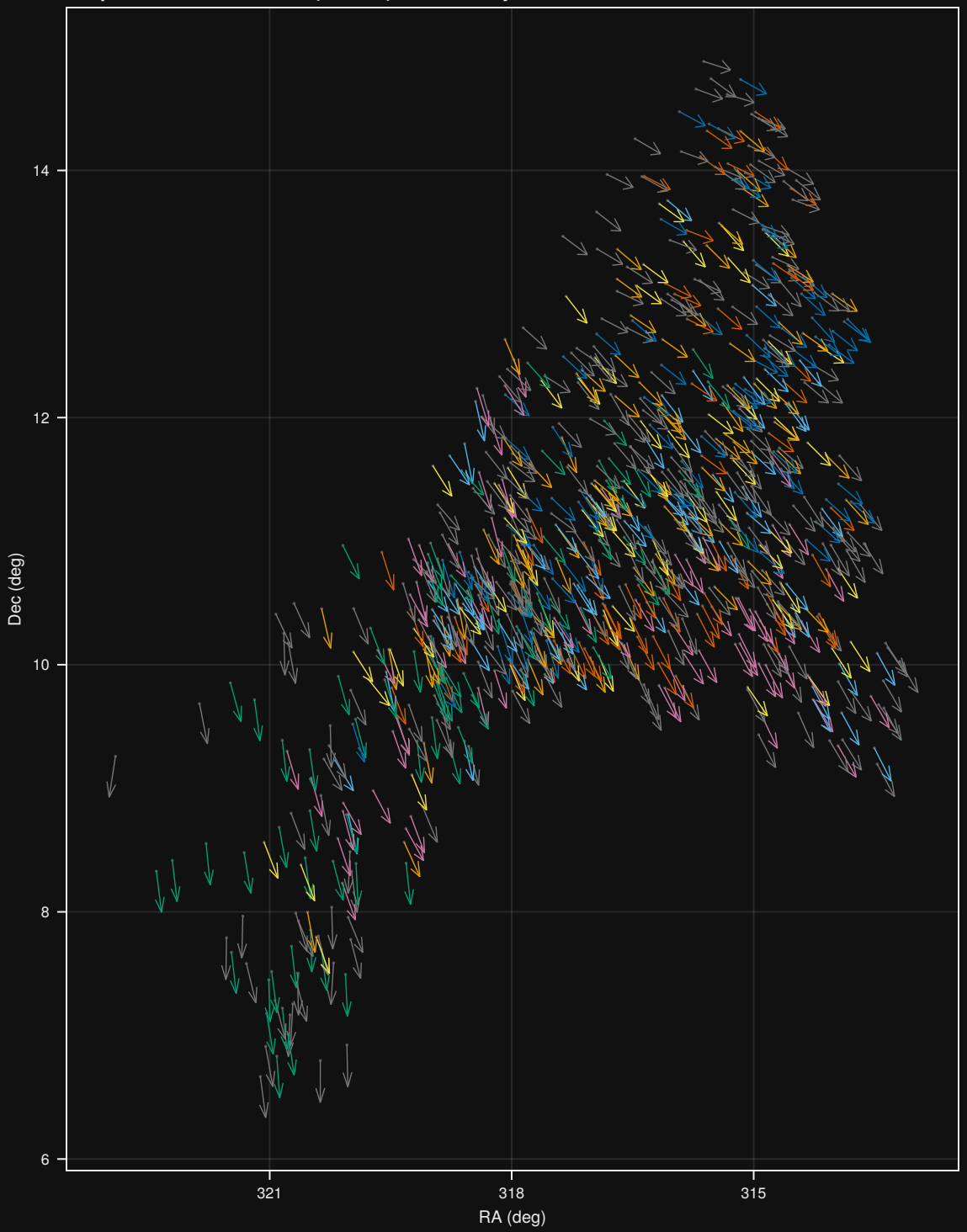


New-21

PM clustering · HDBSCAN · 49 cluster(s) · noise 35%



Proper-motion directions (arrows) coloured by cluster



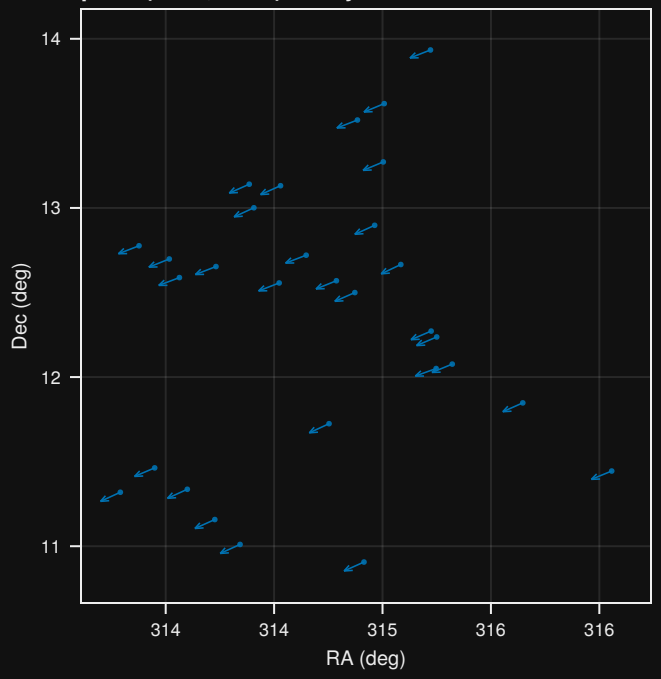
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	265	-2.95	-2.57	1.52	-1.29	0
C1	30	-4.84	-2.71	0.37	-1.17	0
C2	26	-3.37	-2.15	0.20	-1.05	0
C3	26	-0.42	-2.36	0.11	-0.91	0
C4	23	-3.09	-3.91	0.09	-1.09	0
C5	19	-2.66	-2.90	0.14	-1.45	0
C6	19	-2.43	-3.26	0.13	-1.13	0
C7	18	-2.85	-2.28	0.10	-1.25	0
C8	13	-2.95	-1.71	0.22	-1.11	0

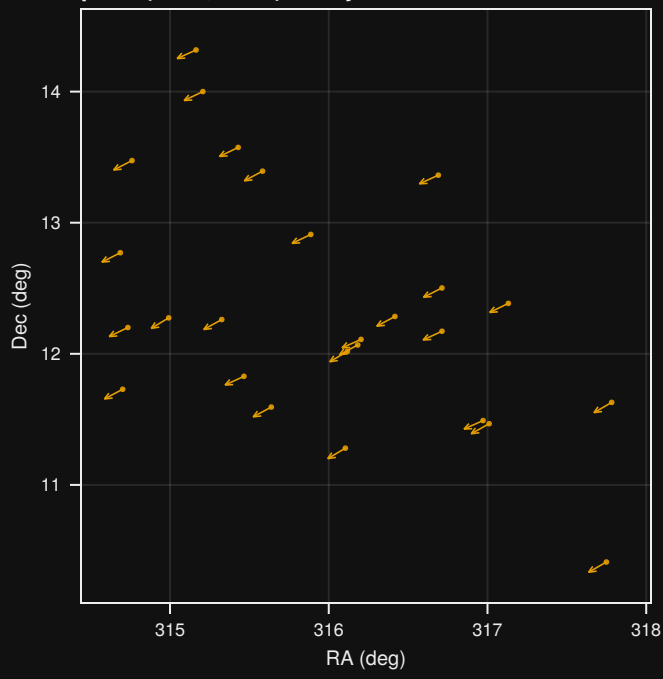
New-21

Per-cluster proper-motion vector maps

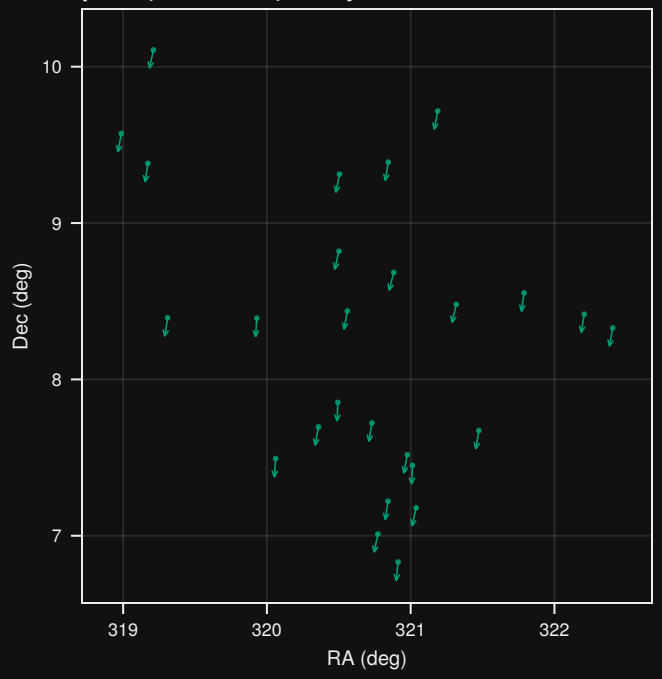
Cluster C1 · n = 30
pm = (-4.84, -2.71) mas/yr



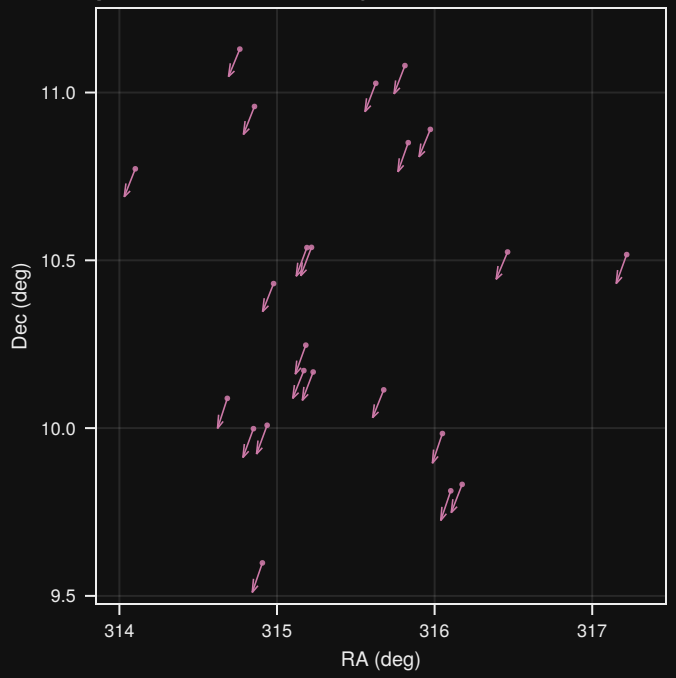
Cluster C2 · n = 26
pm = (-3.37, -2.15) mas/yr



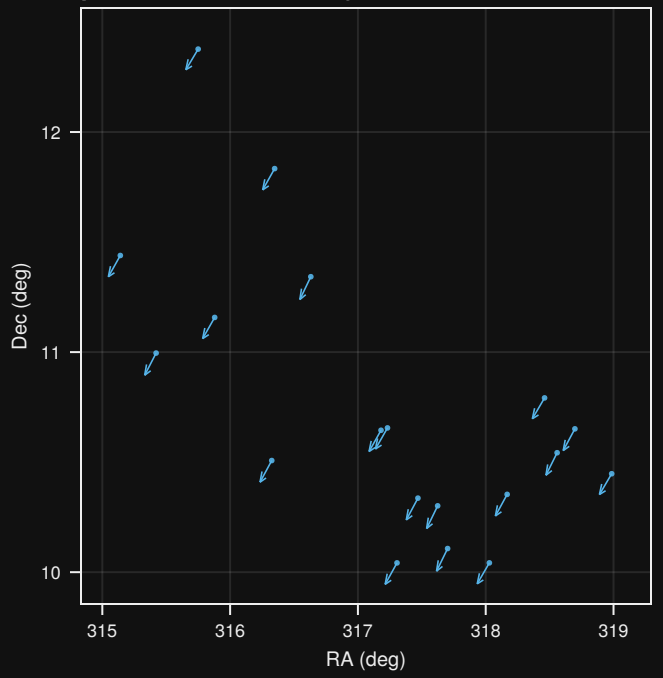
Cluster C3 · n = 26
pm = (-0.42, -2.36) mas/yr



Cluster C4 · n = 23
pm = (-3.09, -3.91) mas/yr



Cluster C5 · n = 19
pm = (-2.66, -2.90) mas/yr



Noise · n = 265

